Strategic Decision Proposal: Data-Driven Selection of an Offshore Clinical Data Delivery Center for XYZ Clinical Data Solutions

1. Introduction

Purpose of proposal: This proposal, prepared for the leadership of XYZ Clinical Data Solutions, outlines a data-driven approach to identifying a shortlist of three candidate countries for the company's first international delivery center. It explains what will be analyzed, why the analysis matters, and how Tableau will be used to compare countries using the OECD Going Digital dataset (OECD, 2025). The proposal does not select a final country; instead, it establishes an analytical framework that will guide the final analysis and country recommendations.

Description of company: XYZ is a hypothetical U.S.-based company that provides clinical data management services to small and mid-sized pharmaceutical, biotechnology, and medical device companies conducting Phase I–IV clinical trials. Its services include building electronic data capture (EDC) systems used to collect trial data, cleaning and validating data, medical coding, and preparing standardized trial datasets for regulatory submission. The company is also expanding its use of AI-assisted tools, including automated detection of data errors. XYZ currently employs approximately 350 staff and supports around 40 active client studies.

Type of expansion: XYZ is considering an offshore expansion by establishing a clinical data delivery center with 50–75 employees in another country. The center would initially handle data cleaning, query management, and medical coding, while client-facing project management and regulatory submission support would remain in the United States. It would not serve local customers. Its value proposition is to expand delivery capacity, support faster turnaround by extending data-cleaning activities across time zones, and provide access to skilled data professionals who are increasingly scarce and costly in the United States, while maintaining the data quality and security clients expect. Because XYZ manages sensitive and regulated patient data, the location decision must consider not only talent and cost, but also data security, digital infrastructure, and openness to foreign investment.

2. Objective, Assumptions, and Initial Strategy

Objective: The objective of this analysis is to identify a shortlist of three candidate countries that could support XYZ's first offshore clinical data delivery center. Countries in the OECD Going Digital dataset will be evaluated on their readiness to host a regulated, digitally intensive service operation. The goal is not simply to identify the countries with the strongest digital-development scores, but to produce a clear, evidence-based shortlist that reflects XYZ's specific operating needs. These countries will then be examined in greater detail for costs, regulations, and operational fit before a final location decision is made.

Initial strategy: XYZ's initial strategy is to evaluate countries across four strategic pillars, built from selected OECD indicators, that reflect the needs of the proposed center: Talent and Workforce, Trust and Data Security, Business Digital Infrastructure, and Market Openness. Talent will receive the greatest emphasis because the center depends on employees with analytical, technical, and life-science skills. Trust and data security will also be central because XYZ handles sensitive clinical trial data, and a breach could affect the wider business rather than only the new site. Reliable infrastructure is essential for cloud-based data platforms, while market openness helps indicate how easily XYZ can establish a foreign-owned entity and deliver services across borders. XYZ is open to both mature and emerging digital economies, since a lower-cost country with strong talent and adequate data safeguards may prove more attractive than a higher-cost country with only marginally stronger indicators. As a service expansion, the center is not expected to generate new revenue directly; its value will come from increased capacity, faster turnaround, cost efficiency, and stronger client retention. The strategy therefore also considers synergies with XYZ's U.S. teams, including time-zone handoffs that allow work to continue overnight, the time needed to reach full productivity, and the ability to maintain consistent quality standards across both sites.

Assumptions and constraints: Several assumptions and constraints will guide the analysis. The United States will be excluded as XYZ's home market, and the initial entry model assumes a wholly owned center rather than outsourcing, although a partnership may be considered if foreign-investment restrictions prove significant. Because the center will not serve local customers, local market size and saturation are not relevant to this decision. The OECD dataset provides standardized measures of digital readiness but does not capture several factors important to an offshore-services decision, including labor costs, English-language proficiency, time-zone compatibility, political stability, and the depth of the local clinical-research workforce; these will be addressed through supplementary data or clearly stated mock assumptions where appropriate. Preliminary exploration also indicates that indicator coverage varies across countries and is particularly limited for some Trust and Security measures. Countries lacking data for an entire strategic pillar will therefore be treated as data-sparse candidates and assessed separately, rather than excluded or scored poorly, so that missing evidence is not confused with weak performance.

3. Strategic Importance

Importance for XYZ: International expansion exposes firms to significant risk when decisions are made without adequate preparation, readiness, and evidence. For XYZ, the consequences of a poor location choice would extend well beyond the new site. A delivery center that cannot recruit and retain qualified staff, meet sponsor quality expectations, or protect sensitive patient data could damage client relationships, delay trial timelines, and expose the company to regulatory and reputational risk across its entire business. Because the center is intended to strengthen service to existing clients rather than generate new revenue, its success depends on choosing a location that can deliver consistent quality from the start. A structured, data-driven analysis is therefore essential before XYZ commits capital, staff, and client work to a new country.

Reducing uncertainty, identifying attractive locations, and supporting investment decisions: The proposed analysis will reduce uncertainty in three ways. First, it will replace general impressions about popular outsourcing destinations with comparable evidence across all candidate countries in the OECD dataset. Second, it will make XYZ's priorities explicit by linking each strategic pillar to a specific operating requirement, so that countries are judged against what the center actually needs rather than general digital performance. Third, it will identify potential risks and evidence gaps early, before financial and contractual commitments are made. The analysis will also help identify attractive locations, the equivalent of attractive markets for a service delivery center, by showing not only which countries perform well overall, but why they perform well and where their weaknesses lie, and by highlighting countries that perform consistently across all four pillars. This directly supports investment decisions: narrowing the field to three countries allows XYZ to focus the cost of detailed due diligence where it is most likely to pay off, while understanding each country's gaps helps leadership anticipate where additional investment may be required, such as stronger security controls, training programs, or local partnerships.

Role of Tableau in evidence-based decisions: Tableau is well suited to this decision because it involves many indicators, many countries, and trade-offs that leadership will want to examine for themselves. Interactive dashboards will allow decision-makers to compare countries side by side, view performance across the four strategic pillars, and drill down from an overall result to the underlying indicators. Maps and heat maps will make regional patterns and data gaps immediately visible, while filters will allow users to focus on specific pillars, regions, or country groups. Users will also be able to compare countries under different pillar priorities, showing whether a country remains attractive when strategic emphasis changes. This transparency builds confidence in the final recommendation, because leadership can see the evidence behind each shortlisted country rather than relying on a single score or a consultant's judgment alone. The dashboard will also remain useful beyond this decision, since it can be updated as new OECD data becomes available and reused if XYZ later considers a second international site.

4. Key Strategic Decisions

The proposed analysis will address three strategic decisions. Rather than selecting countries in advance, each decision relies on the selected indicators to compare and rank-order countries objectively.

Decision 1: Which indicators best define a country's readiness to host XYZ's delivery center? Candidate indicators from the OECD dataset will be evaluated for their relevance to the center's operating needs, their country coverage, the recency of their data, and their overlap with other measures. Selected indicators will be organized into the four strategic pillars: Talent and Workforce, Trust and Data Security, Business Digital Infrastructure, and Market Openness. Indicators with limited coverage but clear relevance, such as health data sharing, will be retained as supporting evidence rather than included in the core comparison.

Decision 2: How should countries be compared, grouped, and ranked? Countries with data across all four pillars will be ranked on their pillar scores and an overall readiness score and compared under different pillar priorities to test whether their position remains stable. Countries will also be grouped by similar strategic profiles, such as high-talent but lower-security markets. Countries missing data for an entire pillar will be assessed separately on the evidence available, so that emerging economies are not overlooked simply because of gaps in the dataset.

Decision 3: Which risks should caution against or remove a country from consideration? The analysis will distinguish between risk flags and evidence gaps. A risk flag reflects weak observed performance, for example a country the OECD classifies among the bottom three on a core Trust or Market Openness indicator. An evidence gap reflects missing data and will prompt supplementary research rather than a penalty. Countries will be excluded only where a defensible benchmark exists; otherwise, weaknesses will be flagged for management review. Political and regulatory risks, assessed through supplementary governance and data-protection data, will also be reviewed for shortlisted countries. Countries combining strong readiness, manageable risks, and sufficient evidence will form the shortlist of three for due diligence.

5. Proposed Methodology

Importing and preparing the dataset: Earlier exploration examined individual indicators and their country coverage; this analysis extends that work by combining selected indicators into strategic pillars, comparing countries across them, and testing results under different priorities. The OECD Going Digital dataset will be imported into Tableau and filtered to candidate countries by removing the United States and aggregate rows such as EU27, G20, and the OECD average, although the OECD average may be retained as a benchmark. The analysis will use the standardized Distance scores, which place indicators measured in different units on a common 0–100 scale. Preliminary checks confirm that higher Distance scores consistently represent more favorable performance, including for privacy-violation and regulatory-restrictiveness indicators. Because some country–indicator pairs appear more than once due to multiple classifications, values will be averaged rather than summed to avoid double-counting. Further preparation details are provided in Appendix B.

Selecting key indicators and performance measures: Candidate indicators will be grouped into the four strategic pillars using Tableau groups, with the proposed indicators listed in Appendix A. Calculated fields will be created for each pillar score, an overall readiness score, data completeness, risk flags, and evidence-gap flags. A country will be considered fully scorable when at least one core indicator is available in every pillar; countries that do not meet this rule will be assessed separately as data-sparse candidates.

Comparing countries and creating visualizations: Countries will be compared using several complementary views:

- a map showing overall readiness geographically,
- a heat map of countries by indicators, which reveals both strengths and data gaps,
- ranked bar charts of pillar and overall scores, and
- a scatter plot comparing Talent against Trust and Data Security, divided into quadrants by average lines to highlight countries strong on both.

Tableau parameters will allow users to change the relative emphasis on each pillar and observe how rankings shift, identifying countries that remain strong under different priorities. Tableau clustering will be explored to group countries with similar strategic profiles, and the dataset's "Fastest changing" classification will be used to highlight countries whose performance is changing most rapidly, interpreted in light of each indicator's direction.

Expected final output: The final output will be an interactive executive dashboard that combines these views with filters and drill-down from pillar scores to individual indicators. Risk flags and evidence gaps will be displayed separately, so users can distinguish weak performance from missing information. The dashboard will support a transparent shortlist of three countries for deeper cost, regulatory, and operational due diligence.

6. Additional Data Requirements

The OECD dataset provides a strong foundation for comparing digital readiness, but several additional types of information would strengthen the analysis. Where real data is unavailable, clearly labeled mock data will be used.

Company-specific: Mock company data would define the scope and constraints of the expansion, including an assumed investment budget of approximately US$4–6 million over the first two years, a target of 50–75 employees reached in phases, and the service lines to be moved offshore first. Information on XYZ's client profile would also be valuable, such as the share of sponsors requiring data to remain within specific regions or requiring on-site audit access, since these requirements could rule out certain locations regardless of their digital readiness. Competitive positioning, such as where other clinical data providers already operate delivery centers, would indicate both proven locations and potential competition for talent.

Country-specific: Several country factors outside the OECD dataset are important to this decision. These include the strength of national data-protection laws, which can be reviewed through UNCTAD's Data Protection and Privacy Legislation Worldwide tracker (UNCTAD, n.d.); political and governance stability, measured by the World Bank Worldwide Governance Indicators (World Bank, n.d.); English-language proficiency, measured by the EF English Proficiency Index (EF Education First, n.d.); and time-zone overlap with U.S. business hours. Labor market conditions, particularly the availability of life-science graduates and professionals with clinical research experience, would also complement the OECD talent indicators.

Operational: Labor costs for data management and technical staff are the most significant gap, because cost efficiency is a central reason for the expansion. Because reliable salary data for these specific roles is not readily available across countries, labor costs will likely be represented using mock estimates informed by published sources such as ILOSTAT (International Labour Organization, n.d.). Office and setup costs, and potential local partners such as universities, staffing firms, or established clinical research organizations, would help estimate how quickly the center could reach full productivity. Physical logistics and supply chain networks are less relevant, because the center will deliver digital services rather than physical goods.

External datasets: Two external datasets would be particularly useful. First, the Global AI Index would help assess each country's AI ecosystem (Tortoise Media, n.d.), which is relevant as XYZ expands its AI-assisted data review services. Second, ClinicalTrials.gov would provide the number of registered clinical trials per country (National Library of Medicine, n.d.), indicating the depth of each country's clinical-research activity and its pool of experienced staff.

To keep the analysis focused, labor costs, English-language proficiency, and clinical-research activity will be prioritized for integration. These will be compiled into a single supplementary dataset and joined to the OECD data by country in Tableau, focusing on the most promising and data-sparse candidates rather than all countries. This supplementary evidence will be especially valuable for data-sparse countries, where it can help fill gaps in the OECD indicators.

7. Conclusion

Importance of data-driven decision-making: Choosing the location for XYZ's first offshore delivery center is a long-term commitment. For a company entrusted with sensitive clinical trial data, the consequences of a poor choice would extend well beyond cost. A data-driven approach reduces this risk by grounding the decision in comparable, transparent information that leadership can examine and question for themselves.

How the methodology supports a rational decision: The proposed methodology links each stage of the analysis directly to XYZ's operating needs. Countries will be evaluated across four strategic pillars that reflect what the center requires to succeed: talent, data security, digital infrastructure, and market openness. By separating weak performance from missing evidence, testing whether countries remain strong under different priorities, and flagging risks early, the analysis ensures that promising countries are not overlooked and that weaker ones are not advanced on incomplete information. Supplementary data on labor costs, language proficiency, and clinical-research activity will add the business context that the OECD indicators alone cannot provide.

Foundation for the final analysis: This proposal does not select a final location. Instead, it establishes the framework for the next phase of work, in which the Tableau dashboard will be built, countries will be compared in depth, and a shortlist of three candidate countries will be recommended for detailed cost, regulatory, and operational due diligence. The result will be an expansion decision that is strategic, evidence-based, and aligned with XYZ's commitment to data quality and security.

References

EF Education First. (n.d.). *EF English Proficiency Index*. Retrieved September 19, 2026, from https://www.ef.com/epi

International Labour Organization. (n.d.). *ILOSTAT*. Retrieved September 19, 2026, from https://ilostat.ilo.org

National Library of Medicine. (n.d.). *ClinicalTrials.gov*. Retrieved September 19, 2026, from https://clinicaltrials.gov

OECD. (2025). *Going Digital Toolkit* [Data set]. Retrieved September 19, 2026, from https://goingdigital.oecd.org

Tortoise Media. (n.d.). *The Global AI Index* [Data set]. Kaggle. Retrieved September 19, 2026, from https://www.kaggle.com/datasets/katerynameleshenko/ai-index

UNCTAD. (n.d.). *Data protection and privacy legislation worldwide*. Retrieved September 19, 2026, from https://unctad.org/page/data-protection-and-privacy-legislation-worldwide

World Bank. (n.d.). *Worldwide Governance Indicators*. Retrieved September 19, 2026, from https://www.worldbank.org/en/publication/worldwide-governance-indicators

Appendix A. Proposed Indicators by Strategic Pillar

Coverage shows the number of candidate countries (out of 43) with a Distance score for each indicator, based on preliminary exploration of the OECD Going Digital dataset.

Core Indicators

Strategic pillar	Indicator (OECD)	OECD policy dimension	Coverage	Relevance to XYZ
Talent and Workforce	New tertiary graduates in science, technology, engineering and mathematics as a share of new graduates	Jobs	42	Future pipeline of analytical and technical staff
	Digital-intensive sectors' share in total employment	Jobs	42	Workforce already experienced in digital work
Trust and Data Security	Share of enterprises in which own employees carry out ICT security related activities	Trust	25	In-house cybersecurity maturity for protecting clinical data
	Share of Internet users experiencing abuse of personal information or privacy violations	Trust	32	Prevalence of privacy risks experienced by internet users
Business Digital Infrastructure	Share of businesses with broadband contracted speed of 30 Mbps or more	Access	30	Reliable access to cloud-based data platforms
	Share of businesses purchasing cloud services	Use	36	Business familiarity with cloud-based workflows
Market Openness	OECD Foreign Direct Investment Regulatory Restrictiveness Index	Market openness	43	Ease of establishing a foreign-owned entity
	OECD Digital Services Trade Restrictiveness Index	Market openness	42	Ease of delivering digital services across borders
	Digitally-deliverable services as a share of commercial services trade	Market openness	43	Existing experience in exporting digital services

Supplementary Indicators (supporting evidence, not included in core scores)

Indicator (OECD)	OECD policy dimension	Coverage	Reason for supplementary role
Share of ICT task-intensive jobs	Jobs	29	Relevant to talent, but limited coverage
Workers receiving employment-based training as a share of total employment	Jobs	28	Indirect link to immediate hiring needs; limited coverage
Health data sharing intensity	Trust	21	Highly relevant to clinical data, but covers only half of candidate countries

Appendix B. Data Preparation and Scoring Framework

Component	Proposed approach	Purpose
Candidate countries	Exclude the United States (home market) and aggregate rows (EU27, G20, OECD average)	Keep rankings limited to 43 candidate countries
Benchmark	OECD average may be retained as a reference line	Provide a point of comparison
Measure type	Filter to Type = "Distance"	Place all indicators on a common 0–100 scale
Score direction	Confirmed: higher Distance = more favorable performance, including for restrictiveness and privacy-violation indicators	Ensure consistent interpretation
Duplicate rows	Aggregate using AVG rather than SUM	Prevent double counting where a country has multiple classifications
Pillar assignment	Group selected indicators into four pillars using Tableau groups	Organize the analytical model
Pillar score	Average of available core indicators within each pillar	Summarize performance by pillar
Overall readiness score	Combination of the four pillar scores, with adjustable pillar emphasis through Tableau parameters	Enable overall comparison and sensitivity testing
Fully scorable country	At least one core indicator available in every pillar	Define eligibility for the overall score
Data-sparse country	No core indicators available in one or more pillars	Retain for separate assessment rather than exclude
Data completeness	Percentage of the nine core indicators available for each country	Show the strength of the underlying evidence
Risk flag	Country classified among the bottom three on a core Trust or Market Openness indicator	Identify weak observed performance
Evidence-gap flag	Missing data for an entire pillar	Prompt supplementary research rather than a penalty
Supplementary data	Labor costs, English proficiency, and clinical-research activity joined by country	Add business context not captured by OECD data